

Exam Date & Time: 19-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester Makeup Examination, June -2025

Introduction to Internet of Things [NPT-3202]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A corporate office relies on traditional internet-connected devices like desktop computers, printers, VoIP phones, and Wi-Fi routers. Employees manually configure these devices, and IT staff handle maintenance, security updates, and troubleshooting. (5)
 - A) How does IoT differ from traditional internet-connected devices?
 - B) How does security and privacy impact IoT devices, and what measures can be taken to protect them? Explain your answer with scenario. (5)
- 2) Build a code to develop home automation system using Arduino. (5)
 - A)
 - B) Explain the procedure to connect Bluetooth (HC-05) or WiFi (ESP8266) with Arduino. (5)
- 3) How do IoT protocols differ in terms of power consumption and data rate? (5)
 - A)
 - B) Identify the key security considerations in IoT communication protocols. (5)
- 4) What are the key layers of Sensor Cloud architecture? (5)
 - A)
 - B) A hospital uses IoT-enabled medical devices to monitor patients' vital signs in real time. Due to network limitations, transmitting all data to a remote cloud causes delays in emergency response. How can fog computing enhance patient monitoring and emergency alerts? (5)
- 5) A large cloud service provider operates multiple data centers that handle massive amounts of internet traffic, hosting various services like streaming, cloud storage, and enterprise applications. Analyse how SDN used in such data centers. (5)
 - A)
 - B) Explain the challenges in managing large-scale IoT data. (5)

-----End-----